EvoDrive: Pareto Evolution for Safety-Critical Autonomous Driving via Self-Improving LLM Agents

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HarnessRSI 65 ★★

evidence: improver

Paper

1 What changes, what is kept

1 Propose a bounded generator edit
Parameters · operator graph · scene policy
Reusable mechanisms; ego policy stays fixed

2 Review, repair, validate, compile
Critics advise; deterministic gates fail closed
Illegal edits never reach the simulator

3 Route: select / reject / audit
World evaluator predicts attack + realism
Uncertain cases can force simulator audits

4 Measure in closed-loop simulation
MetaDrive: collisions + realism diagnostics
CARLA: CR(1 − OS) + validity-weighted ARS

5 Keep trade-offs, update memory
Archive: nondominated simulator-labeled points
Active child: frontier / bounded trade-off gates

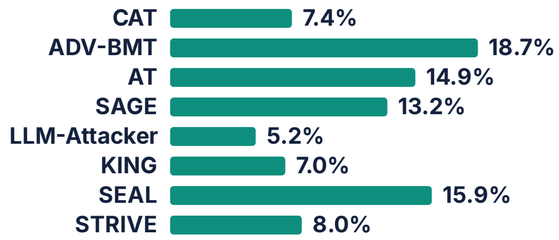
Nested improvement: evolve the router
Earlier simulator labels → judge → later routing
Cannot rewrite simulator labels or archive rules

MetaDrive protocol: 240 evolution scenes; 100 protected test scenes. Select active generator + ≤3 descendants on evolution pool, then freeze.

10 epochs × 20 proposal slots. R ≥ 0.95, tolerance 0.05; default-tuning split not stated.

2 Frontier gain, not safer driving

PF-Area@3 · MetaDrive · Table 1 · higher is better



IDM/RL means; ≤3 descendants ≠ equal total compute vs parents. No variance/CI or run count.

4 Where gains are absent / concentrated

- **Policy dependence:** ADV-BMT PF gain 33.4% RL vs 4.0% IDM. Zero: LLM-Attacker/IDM, KING/RL, STRIVE/RL, SAGE w=0.50/RL (Table 11).
- **Trade-off:** CAT attack 0.345 → 0.330; RP improves 10.66% → 3.85%.
- **SAC, route-disjoint test:** avg CR 0.163 → 0.106. Gains: straight/turning obstacles + right-turn (0.350 → 0.000).
- **But vehicle-passing CR worsens 0.050 → 0.200;** unprotected-left stays 0.500; Lane Change, Red-light and Crossing stay at zero. Table 6: no run count/CI.

3 Controls that matter

Three independent evolution runs

ADV-BMT / RL only · Table 4 · PF-Area@3 (%)

Variant	Reported mean ±
Full EvoDrive	**31.0 ± 4.2**
No critic / repair	26.8 ± 3.8
No long-term memory	19.1 ± 3.5
Non-LLM Pareto	10.6 ± 2.8
Actor-only, LLM calls matched	21.2 ± 4.0

± type unspecified; full / no-critic ranges overlap. Non-LLM budget matching not stated.

Does evolving the evaluator help? · Table 5

Same candidates + sim budget	Static	Evolving
Norm. evals to 90% HV ↓	0.60 ± 0.04	0.51 ± 0.03
Norm. top-10 yield ↑	1.30 ± 0.06	1.44 ± 0.05

Updates use earlier-round labels. Run count / ± type not stated for Table 5.

Held-out naturalistic check · Table 2

Motion JSD 0.073 → 0.067; interaction JSD 0.091 → 0.081; OOS 7.4% → 6.0%. Evaluation only, not selection feedback.

Transfer / cost / missing controls

Frozen SAC-selected scenarios: BasicAgent attack 0.0612 → 0.1434 (Table 17).

CARLA run: 232 calls, 13.5 h, $5.88 API estimate. Baseline tuning budgets unreported; CARLA generator test/evolution disjointness not explicit.

Takeaway

Typed generator edits + simulator-grounded Pareto retention improve attack–realism trade-offs. Evolving the router helps at matched simulation budget. Uneven downstream gains are not deployment-safety guarantees.

Authors' limit: simulation-only, fixed policies; unseen policies/platforms and real-world failures may be missed. Hardware-in-loop is future work.

Our read · HarnessRSI

Score 65 · ★★ · evidence: improver

Reports: MetaDrive test; repeated ablation; matched-budget router; cross-policy transfer.

Does not establish: compute parity, headline CIs, leakage audit, real-world safety. See judgment note in SOURCE.